Ruben Otano

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github.com/RubenOtano | linkedin.com/in/rubenotano | AI/ML Engineer

Education

- B.S. in Computer Science from Florida International University — Expected 2028
  Minor in Deep Learning & Mathematics
- A.A. in Computer Science from Miami Dade College — Graduated Jun. 2026 | GPA 4.00

Work History

AI/ML Engineer at CodeCrunch, Miami, FL *Sept. 2025 to Present*

- Analyzed datasets of thousands of users for a modern ed-tech startup, building data pipelines that segmented and enriched users to power CodeCrunch's reach-out program and marketing campaigns.
- Built agentic AI workflows that automated large-scale research, gathering and synthesizing information on thousands of users so outreach could be personalized without manual work.
- Generated personalized outreach and marketing from the analyzed data, increasing engagement across campaigns while keeping a small team's workload low.
- Worked closely with the founding team to ship features quickly, turning raw user data into measurable growth for the platform.

Projects

NeuraScan — Chest X-ray Pathology Detection *2026*

- Built a multi-label thoracic disease classifier around a DenseNet-121 backbone in PyTorch with Grad-CAM localization so findings were highlighted for clinician review.
- Shipped an interactive in-browser demo in Next.js and React that runs the scan, overlays heatmaps, and reports per-pathology confidence in real time.

OncoSeg — Brain Tumor MRI Segmentation *2026*

- Implemented a 3D U-Net (nnU-Net style) to segment glioma sub-regions across multi-modal MRI, evaluating with volumetric Dice so tumor regions were measured accurately.
- Created a canvas-rendered slice viewer with a live segmentation overlay and tumor-volume readout so users could scrub through the scan.

CardioPulse — Real-Time ECG Arrhythmia Detection *2026*

- Built a 1D residual CNN with a temporal transformer head to classify arrhythmias like AFib and PVC from single-lead ECG, targeting low-latency on-device inference.
- Developed a live streaming ECG visualizer with beat-level alerting and heart-rate readout so abnormal rhythms surfaced as they happened.

ClinicalLM — Retrieval-Augmented Clinical Assistant *2026*

- Built a grounded medical Q&A assistant using hybrid BM25 and dense retrieval over clinical literature, with citation-faithful responses and abstention on weak evidence.
- Implemented a token-streaming chat interface with inline source citations so answers stayed traceable to their sources.

MolForge — Molecular Property Prediction *2026*

- Implemented a directed message-passing graph neural network with PyTorch Geometric and RDKit to predict ADMET properties and drug-likeness for virtual screening.
- Built an animated molecular-graph viewer that renders 3D structure and live property predictions so candidates could be compared quickly.

VitalStream — ICU Deterioration Early-Warning *2026*

- Designed a temporal-fusion model that forecasts patient deterioration from streaming ICU vitals up to 6 hours ahead, with feature attributions per vital sign.
- Built a real-time dashboard with live vitals sparklines, a risk gauge, and an alert and event log so care teams could act early.